CONTEXTUAL WATER RISK ASSESSMENT SYSTEM (CW-RAS)

MINI PROJECT REPORT

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COLLEGE OF ENGINEERING PERUMON

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This is to certify that the report entitled **CONTEXTUAL WATER RISK ASSESSMENT SYSTEM (CW-RAS)** submitted by **HRITIK KRISHNA** (PRN23CS071), **IVAN JOHN BENNY** (PRN22CS074), **NANDU M V** (PRN22CS099) & **SRUTHI S** (PRN22CS121) to the APJ Abdul Kalam Technological University in partial fulfillment of the B.Tech. degree in Computer Science and Engineering during the academic year **2025 - 26** is a bonafide record of the project work carried out by them under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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DECLARATION

We hereby declare that the project report **CONTEXTUAL WATER RISK ASSESSMENT SYSTEM (CW-RAS)**, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under supervision of Mrs. SIJIMOL K .

This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources.

We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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ABSTRACT

Water-related risks such as floods and droughts pose significant challenges to environmental sustainability and human livelihoods, especially at the local administrative level. Accurate and accessible risk assessment tools are essential for effective planning and mitigation. This project presents the development of a Contextual Water Risk Assessment System (CW-RAS), a data-driven web-based application designed to evaluate flood and water scarcity risks at the panchayat level.

The system integrates hydrological and environmental data collected from reliable government sources, including the Water Resources Information System Kerala and the Central Ground Water Board. These datasets are consolidated into a unified master dataset to ensure consistency and efficient processing. Key parameters considered in the analysis include rainfall deviation, groundwater level fluctuations, and land-use characteristics such as urbanization and forest cover.

The core of the system is the SR-SA v2 (Simplified Risk-Score Aggregation) algorithm, which computes individual risk components and combines them using a weighted aggregation model. Rainfall and groundwater parameters are assigned higher weights, while land-use contributes contextually to the final score. Additionally, a Surface Water Moderation Factor is incorporated to adjust water scarcity risk based on proximity to water bodies.

The system generates a composite risk score and classifies it into qualitative categories such as low, moderate, and high risk, enhancing interpretability for users. The application is deployed using modern web technologies and hosted on a scalable platform, enabling easy access and real-time analysis.

This project demonstrates a transparent, computationally efficient, and context-aware approach to water risk assessment, providing a practical tool for decision-making at the local level.

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CHAPTER 1

INTRODUCTION

Water is one of the most critical natural resources, essential for sustaining life, agriculture, and economic development. However, increasing climate variability and human activities have led to a rise in water-related risks such as floods and droughts. These risks significantly affect local communities, particularly at the panchayat level, where planning and resource management are often limited by the lack of accessible and localized analytical tools.

Floods are primarily caused by excessive rainfall, poor drainage systems, and land-use changes such as urbanization, which increase surface runoff. On the other hand, droughts or water scarcity conditions arise due to insufficient rainfall, declining groundwater levels, and inadequate water conservation practices. Traditional methods of assessing these risks often rely on large-scale models or manual analysis, which may not provide accurate or timely insights at smaller administrative levels.

With the increasing availability of open hydrological data from platforms such as the Water Resources Information System Kerala and the Central Ground Water Board, there is an opportunity to develop systems that can utilize this data for localized risk assessment. However, integrating and interpreting such diverse datasets remains a challenge.

The Contextual Water Risk Assessment System (CW-RAS) is designed to address this gap by providing a simple, transparent, and data-driven approach to evaluate water-related risks. The system combines multiple parameters, including rainfall deviation, groundwater fluctuation, and land-use characteristics, to compute a composite risk score for a given location. By focusing on contextual and location-specific factors, the system enables more accurate and meaningful risk analysis.

The project also emphasizes interpretability and usability. Instead of relying on complex predictive models, it uses a deterministic scoring approach through the SR-SA v2 algorithm, making it easier to understand, implement, and justify. The system is deployed as a web-based application, allowing users to easily input a location and

obtain risk assessments in real time.

Overall, this project aims to provide a practical and efficient tool for assessing flood and drought risks, supporting better decision-making and resource management at the local level.

1.1 Problem Statement

Water-related risks such as floods and droughts are becoming increasingly frequent due to climate variability and changes in land use. These risks have severe impacts on agriculture, infrastructure, and daily life, especially at the local administrative level such as panchayats.

Existing methods for assessing water risks are often large-scale, complex, and not easily accessible to local users. Moreover, available data from sources like the Water Resources Information System Kerala and the Central Ground Water Board are not directly integrated or utilized for localized risk evaluation.

There is a lack of a simple, transparent, and data-driven system that can combine multiple parameters such as rainfall, groundwater levels, and land-use characteristics to provide an accurate assessment of flood and drought risks for a specific location.

Hence, there is a need to develop a system that can integrate these datasets and generate meaningful risk assessments in an easy-to-understand manner for effective local-level decision-making.

1.2 Objective

- To design and develop a Contextual Water Risk Assessment System (CW-RAS) for evaluating flood and water scarcity risks at the panchayat level.
- To collect reliable hydrological and environmental data from sources such as the Water Resources Information System Kerala and the Central Ground Water Board.
- To integrate data from multiple sources into a single master dataset for consistent and efficient analysis.

- To analyze important parameters including rainfall deviation, groundwater level fluctuations, urbanization, and forest cover.
- To assign appropriate weights to different parameters to reflect their influence on flood and drought conditions.
- To compute a composite risk score and classify it into meaningful categories such as low, moderate, and high risk.
- To incorporate contextual factors such as land use and surface water availability for more accurate assessment.
- To develop a web-based application that allows users to input location data and obtain real-time risk analysis.
- To ensure that the system is simple, transparent, and easy to interpret, making it suitable for practical use at the local level.
- To provide a tool that can assist in better planning, resource management, and decision-making related to water risks.

1.3 Purpose and Need

The increasing frequency of floods and droughts due to climate change and human activities has made water risk assessment an important requirement at the local level. Panchayats and local authorities often lack access to simple and effective tools that can help them understand and manage these risks.

Although large amounts of hydrological data are available from sources such as the Water Resources Information System Kerala and the Central Ground Water Board, this data is not directly usable for decision-making without proper analysis and integration. The absence of a unified system makes it difficult to interpret multiple parameters such as rainfall, groundwater levels, and land-use patterns together.

The purpose of this project is to develop a system that can combine these diverse datasets into a single framework and provide a clear assessment of water-related risks.

By using a structured and transparent approach, the system helps in identifying areas that are vulnerable to floods or water scarcity.

There is a strong need for a simple, accessible, and data-driven solution that can provide reliable results without requiring complex technical knowledge. The proposed system fulfills this need by offering an easy-to-use web-based platform that generates meaningful risk insights, thereby supporting better planning, preparedness, and resource management at the local level.

1.4 Organization of Report

This report is organized to provide a comprehensive and logical presentation of the HIV drug discovery project using machine learning. Chapter 1 introduces the global health challenge posed by HIV, the limitations of traditional drug discovery, and the motivation for using machine learning, along with the project's objectives and significance. Chapter 2 reviews existing research on HIV treatments and machine learning applications in drug discovery, identifying gaps addressed by this project. Chapter 3 details the methodology, describing the datasets, machine learning models, data preprocessing, and evaluation techniques used. Chapter 4 covers the implementation and results, comparing different algorithms, presenting findings, and discussing their implications. Finally, Chapter 5 concludes the report by summarizing major contributions, reflecting on limitations, and suggesting directions for future work. This structure guides the reader from background context through detailed technical development to evaluation and conclusions, ensuring clarity and coherence throughout.

CHAPTER 2

LITERATURE REVIEW

2.1 Multi-Index Assessment and Machine Learning Integration for Drought Monitoring Using Google Earth Engine

This research paper presents an advanced drought monitoring system that integrates multisensor remote sensing data with machine learning techniques. The study focuses on analyzing soil moisture dynamics and drought conditions using data collected from multiple satellite sources such as Sentinel-2, MODIS, and SMAP, along with climate datasets. The system is designed to provide large-scale drought monitoring with improved accuracy by combining different environmental indicators. The use of cloud-based platforms like Google Earth Engine allows efficient processing of vast geospatial datasets, making it suitable for regional and national-level drought assessment.

2.1.1 Methodology

The methodology involves collecting and processing multisource data, including optical, thermal, and hydrological datasets. Various drought indices are computed using satellite-derived parameters such as vegetation health, soil moisture, and temperature variations. These indices are then integrated using machine learning techniques, particularly random forest algorithms, to improve prediction accuracy. Data fusion techniques are applied to combine heterogeneous datasets and reduce inconsistencies. The entire processing is performed on Google Earth Engine, which enables large-scale computation and efficient handling of time-series data. The system also performs feature optimization to identify the most relevant parameters influencing drought conditions.

2.1.2 Results

The methodology involves collecting and processing multisource data, including optical, thermal, and hydrological datasets. Various drought indices are computed using satellite-derived parameters such as vegetation health, soil moisture, and temperature variations. These indices are then integrated using machine learning techniques, particularly random forest algorithms, to improve prediction accuracy. Data fusion techniques are applied to combine heterogeneous datasets and reduce inconsistencies. The entire processing is performed on Google Earth Engine, which enables large-scale computation and efficient handling of time-series data. The system also performs feature optimization to identify the most relevant parameters influencing drought conditions.

The results of the study indicate a significant improvement in drought detection accuracy compared to traditional single-parameter methods. The integration of multiple indices and machine learning techniques allows the system to identify complex drought patterns across different regions and time periods. The approach also reduces errors caused by data variability and improves the reliability of predictions. However, the system requires high computational resources and depends on advanced technologies such as cloud computing and machine learning models, which may limit its accessibility for smaller-scale applications.

2.1.3 Conclusion

The study concludes that combining multisensor data with machine learning techniques provides a powerful and accurate approach for drought monitoring. The integration of multiple indices enhances the system's ability to capture complex environmental conditions. However, the complexity of the system, along with its dependency on advanced tools and expertise, makes it less suitable for simple, user-friendly, and localized applications. This highlights the need for a simplified approach that can still utilize multiple parameters effectively.

2.2 Multiparameter Flood Risk Assessment and Management Planning at High Spatial Resolution in the Region of Attica, Greece

This study [**?**] presents a hybrid modeling framework that combines mechanistic within-host models of HIV infection with machine learning components to simulate and predict treatment outcomes under nucleoside reverse transcriptase inhibitor (NRTI) therapy. While traditional pharmacokinetic/pharmacodynamic (PK/PD) models are useful for understanding antiviral efficacy, they often struggle to capture inter-individual variability and uncertainties in patient adherence or drug metabolism. To address these limitations, the authors integrate machine learning modules for parameter estimation, model correction, and simulation acceleration, enabling a more personalized and computationally efficient approach to modeling HIV dynamics under NRTI treatment.

2.2.1 Methodology

The methodology involves the integration of multiple datasets, including rainfall data, terrain information, land-use patterns, infrastructure details, and environmental conditions. Remote sensing data and field measurements are used to collect relevant information. Geographic Information Systems (GIS) are employed for spatial analysis and visualization of flood risk. Advanced simulation models are applied to analyze water flow, runoff, and flood propagation. The system processes these parameters collectively to generate a comprehensive flood risk assessment at high spatial resolution.

2.2.2 Results

The results demonstrate that the multiparameter approach significantly improves the accuracy and reliability of flood risk assessment. The system is capable of producing detailed flood risk maps that can be used for urban planning and disaster management. The integration of multiple parameters helps in identifying vulnerable regions more effectively. However, the approach requires extensive data processing, high computational power, and specialized expertise, which limits its applicability for small-scale

and user-friendly systems.

2.2.3 Conclusion

The study concludes that multiparameter and high-resolution analysis is highly effective for flood risk assessment and management planning. The integration of diverse datasets enhances the quality of predictions and supports informed decision-making. However, the complexity and resource requirements of the system make it less practical for local-level applications. This indicates the need for a simpler and more accessible approach that can still utilize key parameters for effective risk assessment.

CHAPTER 3

SYSTEM DESIGN

This chapter presents the system design of the Contextual Water Risk Assessment System (CW-RAS). The system is developed to assess flood and water scarcity risks at the panchayat level using a data-driven approach. It integrates multiple environmental parameters such as rainfall deviation, groundwater fluctuation, and land-use characteristics into a unified framework using the SR-SA v2 algorithm.

3.1 System Overview

CW-RAS is a web-based application that allows users to input a location and obtain a comprehensive water risk assessment. The system processes environmental data using a weighted scoring model and generates a risk score along with classification.

—

3.2 System Architecture

The system follows a modular architecture consisting of three main components:

- **Data Layer:** Responsible for collecting and storing data such as rainfall, groundwater levels, and land-use information from various sources.
- **Processing Layer:** Implements the SR-SA v2 algorithm to compute risk scores using weighted aggregation.
- **Presentation Layer:** Provides a user-friendly interface for input and displays results visually.

—

3.3 Design Diagram

The system operates through the following flow:

- User inputs location
- System retrieves relevant data from master dataset
- Algorithm processes parameters
- Risk score is calculated
- Output is displayed with classification

—

3.4 Algorithm (SR-SA v2)

The SR-SA v2 algorithm is used to compute flood and water scarcity risks based on multiple parameters.

3.4.1 Input Parameters

- Rainfall Deviation
- Groundwater Level Change
- Land-use (Urban & Forest Cover)

3.4.2 Rainfall Deviation Calculation

$$Rainfall\ Deviation(\%) = \left| \frac{R_{normal} - R_{current}}{R_{normal}} \right| \times 100 \tag{3.1}$$

3.4.3 Groundwater Score

$$GW\ Score = \min\left(\frac{|GW_{last} - GW_{current}|}{3.0} \times 100, 100 \right) \tag{3.2}$$

3.4.4 Land-use Score

$$Urban = \frac{Urban\%}{100} \times 50 \tag{3.3}$$

$$Forest\ Deficit = \frac{(100 - Forest\%)}{100} \times 50 \tag{3.4}$$

$$Land - use\ Score = \min(Urban + Forest\ Deficit, 100) \tag{3.5}$$

3.4.5 Flood Risk Score

$$Flood\ Risk = (0.4 \times R) + (0.4 \times L) + (0.2 \times GW) \tag{3.6}$$

3.4.6 Water Scarcity Risk

$$Scarcity\ Risk = (0.4 \times R + 0.4 \times GW + 0.2 \times L) \times SWF \tag{3.7}$$

3.4.7 Risk Classification

- Score $< 30 \rightarrow$ Low Risk
- $30 \leq$ Score $< 60 \rightarrow$ Moderate Risk
- Score $\geq 60 \rightarrow$ High Risk

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3.5 User Interface Design

3.5.1 Home Page

The home page allows users to input location and select the type of risk assessment.

—

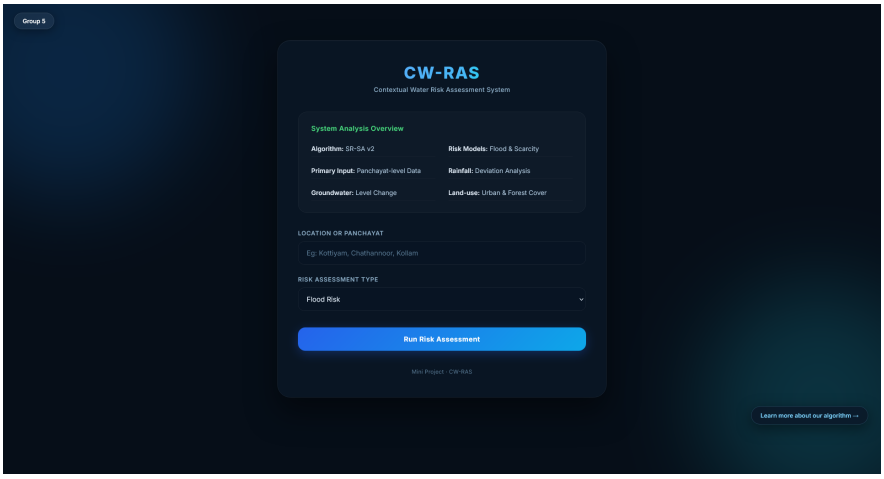

Figure 3.1: Home Page of CW-RAS

3.5.2 Algorithm Page

This page explains the working of the SR-SA v2 algorithm including input parameters and formulas.

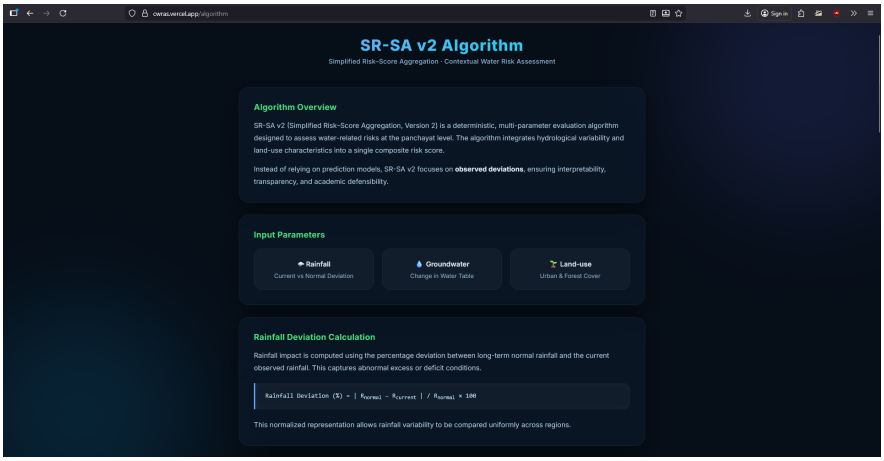

Figure 3.2: Algorithm Overview

—

3.5.3 Risk Calculation and Weight Distribution

This section shows how weights are assigned to different parameters.

—

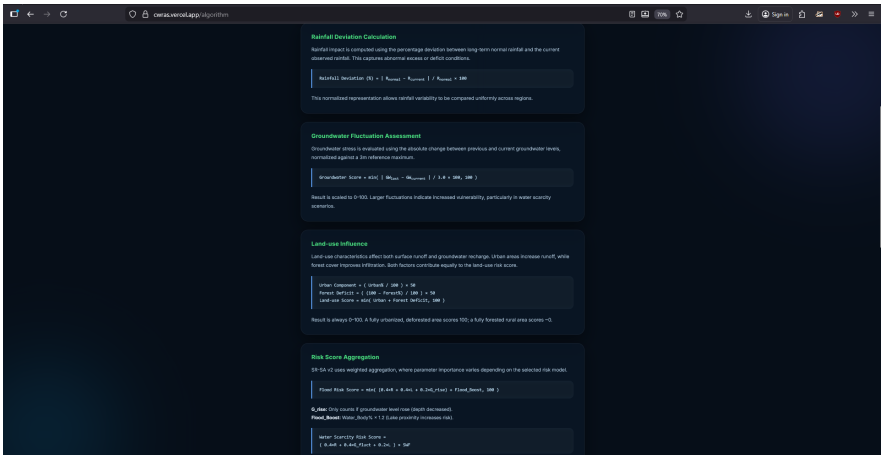

Figure 3.3: Parameter Calculations

3.5.4 Result Page

This page displays the computed risk score, parameter impacts, and classification.

—

3.6 Working of the System

The system starts by accepting user input in the form of a location. Based on this input, relevant environmental data is retrieved from the master dataset. The SR-SA v2 algorithm processes the data by calculating rainfall deviation, groundwater fluctuation, and land-use impact. These parameters are weighted and combined to generate a final risk score.

The score is then classified into low, moderate, or high risk categories. The results are displayed to the user along with detailed parameter breakdowns, enabling better understanding and decision-making.

Figure 3.4: Weight Distribution Visualization

Figure 3.5: Risk Assessment Output

CHAPTER 4

WORK PLAN AND SCHEDULE

This chapter describes the overall work plan and timeline followed during the development of the Contextual Water Risk Assessment System (CW-RAS). The project was carried out in a systematic manner, dividing the work into multiple phases to ensure proper planning, implementation, and testing.

4.1 Work Plan

The development of the CW-RAS system was organized into several stages, each focusing on a specific aspect of the project.

- **Problem Identification:** Understanding the need for a system to assess flood and water scarcity risks at the local level and defining the scope of the project.
- **Data Collection:** Gathering relevant data such as rainfall statistics from WRIS Kerala and groundwater data from CGWB, along with land-use information.
- **Data Integration:** Combining data from multiple sources into a unified master dataset and performing preprocessing such as cleaning and normalization.
- **Algorithm Design:** Designing the SR-SA v2 algorithm to compute risk scores based on rainfall deviation, groundwater fluctuation, and land-use parameters.
- **System Development:** Developing the web-based application including frontend interface and backend processing logic.
- **Testing and Validation:** Testing the system using different input locations to verify accuracy and consistency of results.
- **Deployment:** Deploying the system on Vercel for real-time accessibility.
- **Documentation:** Preparing the project report and related documentation.

4.2 Work Schedule

The project was completed over a period of time following a structured schedule. The major activities and their approximate timeline are shown below.

Table 4.1: Work Schedule of CW-RAS Project

Phase	Activity	Duration
1	Problem Identification and Planning	Week 1
2	Data Collection	Week 2
3	Data Integration and Preprocessing	Week 3
4	Algorithm Design (SR-SA v2)	Week 4
5	System Development	Week 5-6
6	Testing and Validation	Week 7
7	Deployment on Vercel	Week 8
8	Documentation	Week 9

4.3 Summary

The project was executed in a structured and phased manner, ensuring that each stage was completed systematically. This approach helped in efficient development, proper testing, and successful deployment of the CW-RAS system.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The Contextual Water Risk Assessment System (CW-RAS) was successfully designed and developed to evaluate flood and water scarcity risks at the panchayat level using a data-driven approach. The system integrates multiple environmental parameters such as rainfall deviation, groundwater level fluctuations, and land-use characteristics into a unified framework.

The implementation of the SR-SA v2 algorithm enabled efficient computation of risk scores using a weighted aggregation model. The system provides a transparent and interpretable method for assessing water-related risks, avoiding the complexity of traditional models that rely on advanced simulations or machine learning techniques.

By combining data from multiple sources into a master dataset, the system ensures consistency and accuracy in analysis. The web-based interface allows users to easily input location data and obtain real-time results, making the system practical and accessible.

Overall, the project demonstrates a simple, scalable, and effective solution for localized water risk assessment, supporting better decision-making and resource management.

5.2 Future Scope

Although the system provides an effective solution for water risk assessment, there are several areas where it can be further enhanced. At present, the system is limited to the Kollam district due to the availability of localized data, but it can be extended to a larger geographical scope in future implementations.

- **Integration of Real-Time Data:** Future versions of the system can incorporate real-time rainfall and groundwater data for more accurate and dynamic risk assessment.

- **Inclusion of Additional Parameters:** Additional factors such as soil moisture, temperature, drainage systems, and population density can be included to improve accuracy.
- **Use of Machine Learning Models:** Advanced machine learning techniques can be integrated to enhance prediction capabilities and handle complex patterns.
- **GIS and Map-Based Visualization:** Incorporating GIS-based maps can help visualize risk areas geographically for better understanding.
- **Mobile Application Development:** The system can be extended into a mobile application for easier access and wider usage.
- **Automated Data Collection:** APIs can be used to automatically fetch data from sources, reducing manual preprocessing.
- **Decision Support System:** The system can be expanded to provide recommendations and alerts for disaster management.

5.3 Summary

The CW-RAS system lays the foundation for a practical and user-friendly water risk assessment tool. With further enhancements and integration of advanced technologies, it can evolve into a comprehensive decision-support system for managing water-related risks effectively.

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